

TF 21-22(B)

CSE 325-ISD

2105010-Aritra

Question 5

two (2) questions.

5. (a) In Bangladesh, every school teacher is appointed based on MPO regulation. According to this regulation, each candidate has to meet certain qualification criteria defined for each post (of a particular subject) of a certain salary grade. The qualifications are stated in terms of age, subject of graduation, and result at different levels. Example of such requirement is shown below. This is for a particular type of institution, e.g., school, college, etc.

(13)

Table for Q 5(a)

১৩	সহকারী শিক্ষক (সামাজিক বিজ্ঞান)	১) স্বীকৃত বিশ্ববিদ্যালয় হতে সংশ্লিষ্ট বিষয়ে স্নাতক ডিগ্রি / সমমান ও বিভেদ ডিগ্রি / সমমান। অথবা ২) স্বীকৃত বিশ্ববিদ্যালয় হতে সংশ্লিষ্ট বিষয়ে স্নাতক ডিগ্রি / সমমান। সমগ্র শিক্ষাকর্মে ০১ (এক) টি র বেশি ৩য় বিভাগ/শ্রেণি/সমমান গ্রহণযোগ্য হবে না।	অনুষ্ঠ ৩৫ বছর (সম্পদের ইনভেস্টিগেশনের জন্য বয়সসীমা নির্ধারিত)	(১) গ্রেড-১০ (১৬০০০-৩৮৬৪০/-) (২) গ্রেড-১১ (১২৫০০-৩০২৩০/-)
১৪	সহকারী শিক্ষক (পণিত)	১) স্বীকৃত বিশ্ববিদ্যালয় হতে পণিতসহ বিজ্ঞান বিভাগে স্নাতক ডিগ্রি / সমমান ও বিভেদ ডিগ্রি / সমমান। অথবা ২) স্বীকৃত বিশ্ববিদ্যালয় হতে পণিতসহ বিজ্ঞান বিভাগে স্নাতক ডিগ্রি / সমমান। সমগ্র শিক্ষাকর্মে ০১ (এক) টি র বেশি ৩য় বিভাগ/শ্রেণি/সমমান গ্রহণযোগ্য হবে না।	অনুষ্ঠ ৩৫ বছর (সম্পদের ইনভেস্টিগেশনের জন্য বয়সসীমা নির্ধারিত)	(১) গ্রেড-১০ (১৬০০০-৩৮৬৪০/-) (২) গ্রেড-১১ (১২৫০০-৩০২৩০/-)

When a candidate applies for a post, his/her qualifications are matched against such requirements, If the qualification criteria satisfy, he/she is eligible to sit for a recruitment test. Design a BPMN diagram to design the scenario discussed above.

- (b) Draw an ERD to design database for storing the qualification criteria depicted in Q 5(a) for each type of institution.

(12)

- (c) Draw a Mock dashboard (considering scenario of Q 5(a)) to show the list of candidates who are found not eligible to appear at the recruitment test for not satisfying one or more criteria. You have to clearly demonstrate the criteria not matched so that the viewer can easily understand the reason for disqualify.

(10)

Contd P/4

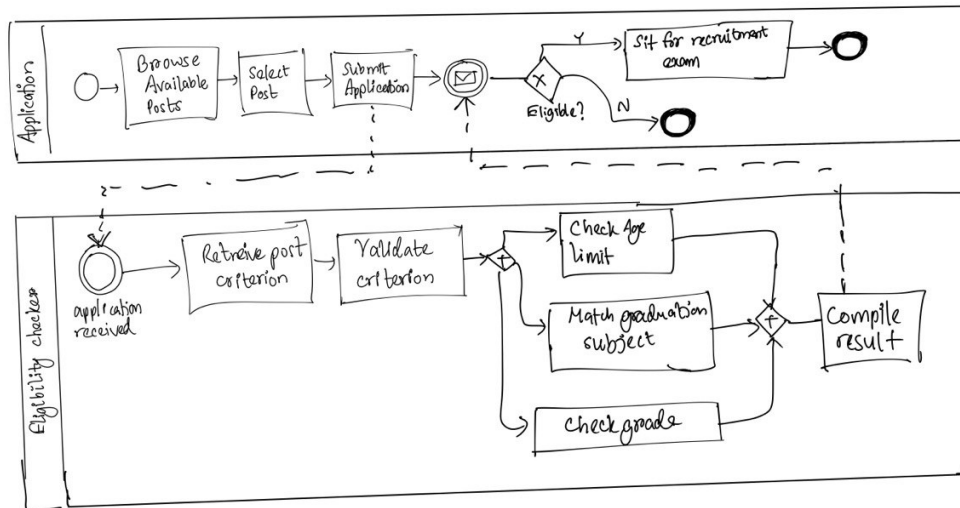


Figure 1: Answer diagram for Question 5(a).

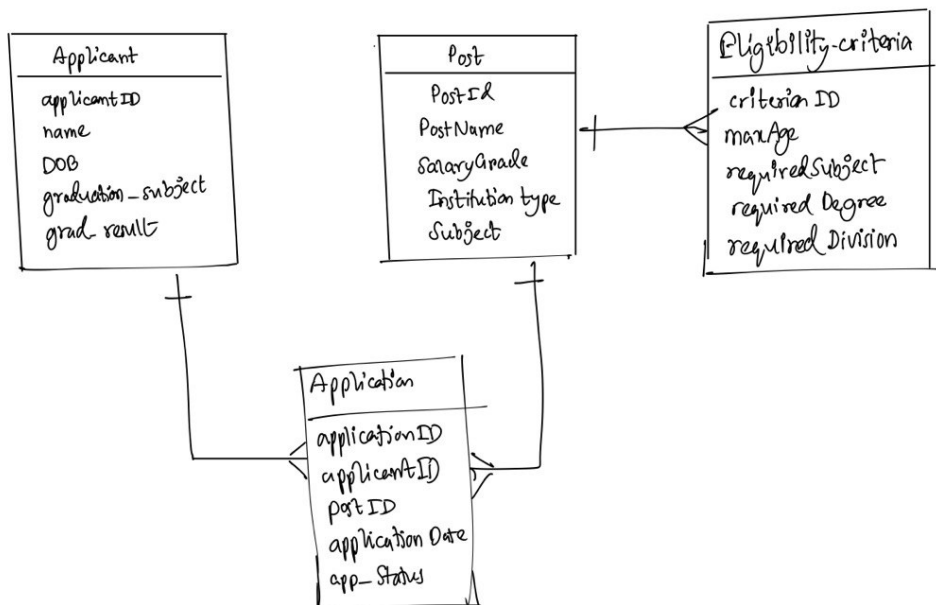


Figure 2: Answer diagram for Question 5(b).

Ineligible candidates				Recruitment cycle — 2024	
Total applicants		Eligible		Not eligible	
38		21		17	
CANDIDATE	APPLIED POST	AGE	SUBJECT	RESULT / GRADE	STATUS
Rahim Uddin	Asst. Teacher (Social Sci.)	× 38 yrs	✓ matched	✓ 2nd class	Ineligible
Nusrat Jahan	Asst. Teacher (Math)	✓ 28 yrs	× unmatched	✓ 1st class	Ineligible
Farhan Hossain	Asst. Teacher (Social Sci.)	× 40 yrs	× unmatched	✓ 1st class	Ineligible
Sadia Islam	Asst. Teacher (Math)	✓ 30 yrs	✓ matched	× 3rd class	Ineligible
Karim Sheikh	Asst. Teacher (Social Sci.)	✓ 25 yrs	× unmatched	× 3rd class	Ineligible
Mitu Akter	Asst. Teacher (Math)	× 37 yrs	✓ matched	× 3rd class	Ineligible
Zahir Chowdhury	Asst. Teacher (Social Sci.)	× 39 yrs	× unmatched	× 3rd class	Ineligible

Figure 3: Answer diagram for Question 5(c).

Question 6(a)

Question: To purchase books, the sites like Amazon.com or Rokomari.com are widely used. Draw a software architecture to implement such a system following Microservice architecture.

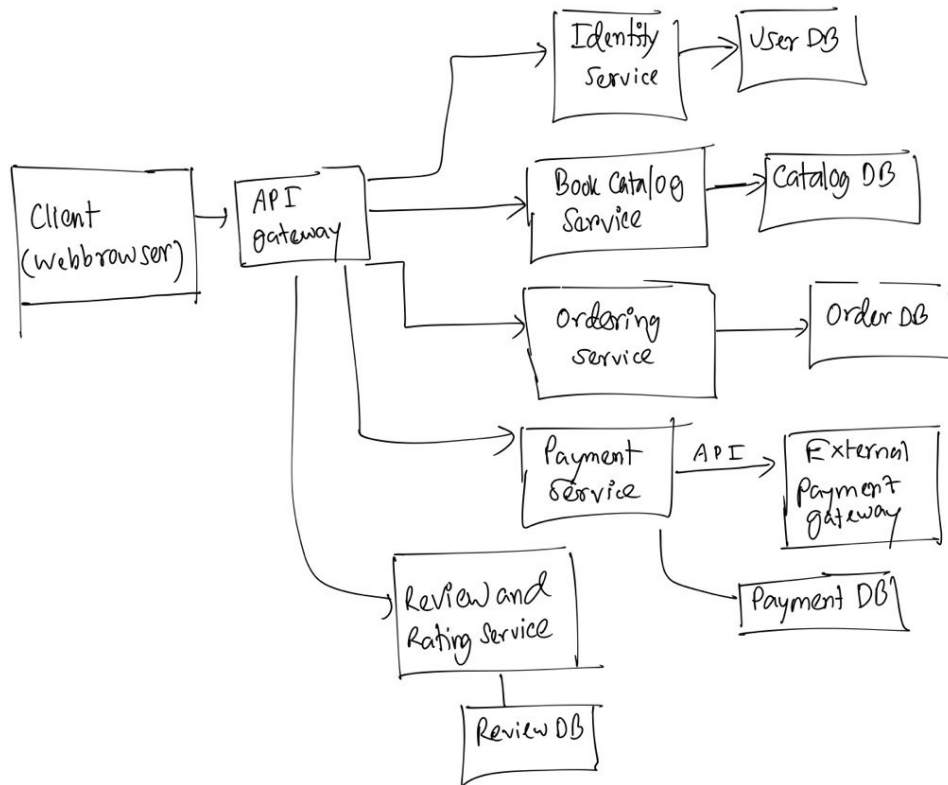


Figure 4: Microservice architecture for an online book-purchasing system.

Question 6(b)

Question: The web portal of stock market experiences frequent write operations and also updated data is expected by viewers. Which cache write strategy is most appropriate for such an application and why?

Answer:

The most appropriate cache write strategy for a stock-market web portal is **Write-Through Caching**.

How it works: Every write operation updates *both* the cache and the underlying database synchronously before the write is acknowledged as successful.

Why it is appropriate here:

- **Strong consistency** — Stock prices, trade volumes, and portfolio values are financial data where even a few seconds of stale information can mislead investors. Write-through guarantees that the cache and database are always in sync, so every read returns the most recently written value.
- **Frequent writes are handled safely** — Because the database is updated on every write, there is no risk of data loss on a cache failure, which is critical for transactional financial records.
- **Fresh data for viewers** — Viewers who query stock prices read directly from the cache. Since the cache is updated on every write, the latency between a price update and a viewer seeing that update is minimised.

Trade-off acknowledged: Write-through introduces slightly higher write latency compared to Write-Behind (write-back), because both the cache and the database must be written before the operation completes. However, for a stock market application, **data accuracy and consistency outweigh write speed**, making this trade-off acceptable.

Question 6(c)

Question: Give three example scenarios where Data-Centered software architecture is appropriate.

Answer:

In a **Data-Centered (Repository) Architecture**, components interact primarily through a shared, central data store rather than directly with each other. It is appropriate when multiple independent components must access or modify a common dataset. Three example scenarios are:

1. **Air Traffic Control Systems.** Multiple independent subsystems—radar processing, weather monitoring, flight scheduling, and ground-to-air communication—all read from and write to a central flight-data repository. No subsystem communicates directly with another; the shared store is the sole source of truth. Data-centered architecture is appropriate here because consistency and real-time accuracy are safety-critical.
2. **Computer-Aided Design (CAD) and Engineering Data Management.** A central design repository holds the master model of a product. Independent tools—stress analysis, thermal simulation, manufacturing planner, and documentation generator—all operate on this shared model without being coupled to each other. Any tool can update the model and all others immediately work with the latest version.
3. **Hospital Information Systems (HIS) / Electronic Health Records (EHR).** A central patient-data repository stores medical history, lab results, prescriptions, and imaging. Separate subsystems—the pharmacy module, the radiology viewer, the billing system, the nursing station—all operate on this shared repository. Data-centered architecture is appropriate here because consistency and a single source of truth for patient data are paramount.

Question 7(a)

Question: There is a common printing service for printing secured official documents in an office which is used by different departments. Sometimes, multiple departments send lots of printing requests simultaneously and want to get assured that their request is being addressed. What technology should be recommended by the designer of such a system? Draw a block diagram for the required design.

Answer:

The recommended technology is a **Message Queue** (e.g., RabbitMQ, Apache ActiveMQ). Each department acts as a *producer* that publishes print requests to a shared queue. The central print service acts as the *consumer*, processing one request at a time in order. This guarantees that:

- No request is lost even if multiple departments submit simultaneously.
- Each department receives an acknowledgement once its message is enqueued, assuring them the request is being addressed.
- The print service is never overwhelmed; it processes at its own pace.

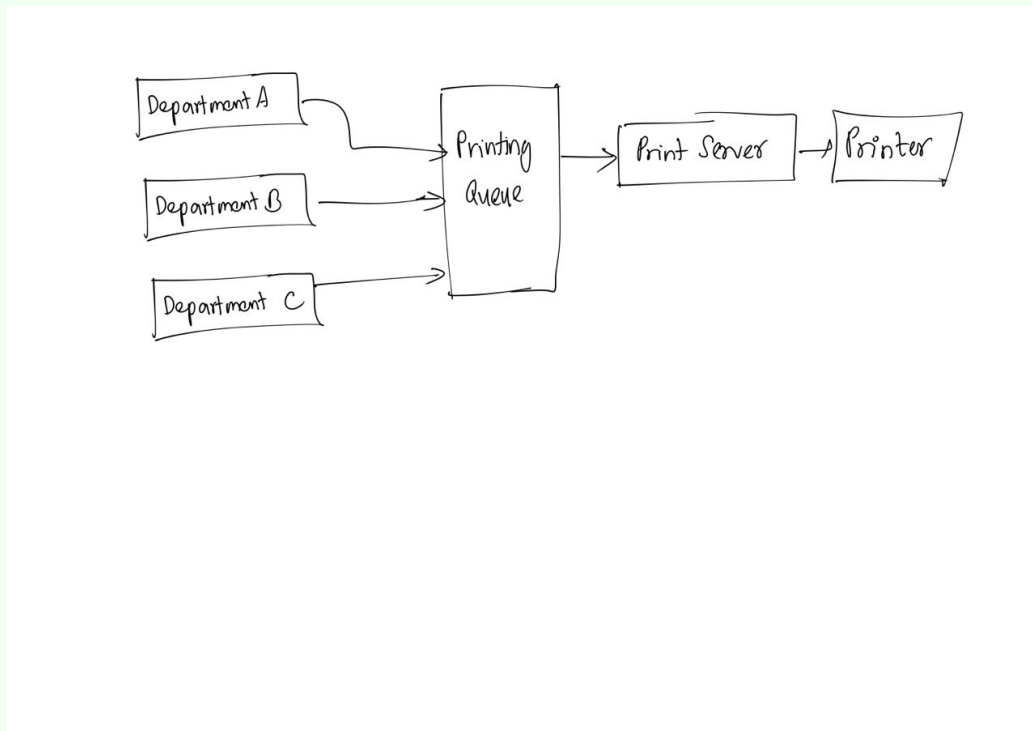


Figure 5: Message Queue based design for the shared printing service.

Question 7(b)

Question: What do you understand by Second Order SQL Injection? Give an example. What is the primary (most recommended) way of preventing SQL injection?

Answer:

Not in syllabus.

Question 7(c)

Question: You have developed an application for different clients. You need to deploy that quickly in different types of servers of the clients. What DevOps technology will serve your purpose?

Answer:

The recommended technology is **Docker** (containerization).

A Docker *image* packages the application together with all its dependencies, libraries, and configuration into a single portable unit. This image can be run as a *container* on any server—regardless of the underlying operating system or hardware configuration—as long as the Docker Engine is installed.

Why it serves the purpose:

- **Platform independence** — the same image runs on Linux, Windows, or cloud VMs without modification.
- **Fast deployment** — a container starts in seconds; no lengthy environment setup on each client server.
- **Consistency** — eliminates the “works on my machine” problem; every client gets an identical runtime environment.

For orchestrating containers across multiple servers, **Kubernetes** can be added on top of Docker to handle scaling and management automatically.

Question 7(d)

Question: In your hardware infrastructure, servers of different capacity are being used. Which load balancing algorithm is best suited for your case?

Answer:

The best suited algorithm is **Weighted Round Robin**.

so basically the other algorithms like least connection, least bandwidth, least response time focus on the connections, not on the capacities of the individual servers. But this weighted round robin can be made to focus on the capacities. If by capacity it meant, how fast the server can respond then least response time might be a solution.

In standard Round Robin, each server receives requests in equal turns regardless of capacity. Weighted Round Robin extends this by assigning a *weight* proportional to each server's capacity—a higher-capacity server receives proportionally more requests per cycle.

Example: If Server A has twice the capacity of Server B, assign weight 2 to A and weight 1 to B. For every 3 incoming requests, A handles 2 and B handles 1, keeping utilisation balanced across unequal hardware.

Question 8(a)

Question: A university department organizes seminars on a number of research topics. These seminars take place in September each year. There is a computer program to help with the administration of these seminars. In the program there is an object representing a seminar, and this Seminar Class has a number of possible states. A seminar is first proposed by a professor. Once dates, times and rooms for the seminar have been arranged, it is considered as scheduled. On the second Monday in August the seminar is advertised and it becomes open for enrolment to the students. When all places are taken, the seminar is full and cannot allow any more enrolment. If a student drops out (cancels his/her application) the seminar reverts to the Open For Enrolment position. Two days before the start of the seminar, enrolment will be closed (no more applicants will be considered). Design a State diagram to show different states of a Seminar in the context discussed above.

Answer:

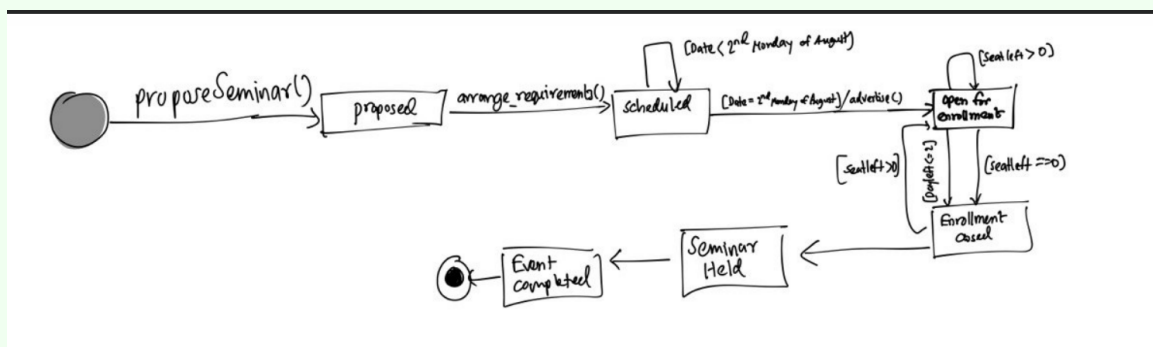


Figure 6: State diagram for the Seminar class.

Question 8(b)

Question: What is the difference between Reflected and DOM-based XSS attack? Draw the flow of a Reflected XSS attack.

Answer:

Not in syllabus.

Question 8(c)

Question: Discuss Interactor, Indirect and Domain viewpoints for the requirement analysis of a Stock Price Prediction system. Mention three non-functional requirements for such a system.

Answer:

Three Non-Functional Requirements:

- **Performance** — The system must generate stock price predictions within 2 seconds of a user query to support real-time decision making.
- **Accuracy** — The prediction model must maintain a minimum accuracy threshold (e.g., $\geq 70\%$) on historical back-testing data.
- **Availability** — The system must be available 99.9% of the time during active market hours to ensure uninterrupted access for users.